2.39 II. Lab Work - Reviewing Python

2.39.1 Work on ones own

1. Enter a three-digit number and find the sum of its digits.
2. Compute formulas:

$$y = -\frac{b + \sqrt{b^2 - 4ac}}{2a}, \quad y = \sin^2(x) + \log_2 |1 + x|. \tag{1}$$

3. "Guess the number"game.
4. Check if a number contains digits 0 or 5.
5. Count four-digit numbers containing digit 0.
6. Function returning sum and product of three numbers.
7. Reverse a string and remove first character using slices.
8. Replace tabs/newlines with spaces and remove punctuation.
9. Remove all max elements from a list (new list).
10. Sort by (1) last digit ascending, (2) descending.
11. Set comprehension for numbers divisible by 3 or 7.
12. Count characters in a string using a dictionary.
13. Find a word starting with capital letter and length < 5 (regex).
14. Extract last name, first name, patronymic from "Full Name: ..."(regex).